

Discovering Short-Term Inefficiencies in High-Frequency Sports Markets via Deep Sequence Modelling

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Abstract

Traditional asset classes, including Equities, Forex, and cryptocurrency, have historically monopolised the discourse surrounding deep machine learning in algorithmic trading. However, these markets are increasingly saturated by institutional High-Frequency Trading firms, diminishing the availability of accessible alpha. Conversely, in-play sports prediction markets remain a highly underexplored frontier for deep sequence modelling. This research indicates that applying predictive algorithms to comprehensive sports data can yield risk-adjusted performance metrics. Specifically, maximum drawdown and equity curve consistency comparable to, or even surpassing, those of conventional quantitative hedge fund strategies. This project looks at shifts in tennis momentum as a complex time-series problem and develops a Context-Aware Gated Recurrent Unit (GRU) to predict 20% declines in implied probabilities. Utilising ATP 2025 tick data, the GRU successfully captured the behavioural inefficiencies of human market participants, yielding an Average Out-of-Sample Return on Investment of +257.10% across a rigorous Walk-Forward Optimisation, while maintaining tightly controlled drawdowns. Ultimately, this report serves as a foundational framework for the future application of deep learning architectures in sports and prediction markets, demonstrating a highly viable, uncorrelated alternative to traditional financial quantitative and algorithmic trading.

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1 Problem Statement

1.1 Introduction and Objective

The contemporary landscape of algorithmic trading is overwhelmingly concentrated on traditional financial asset classes, notably Equities, Foreign Exchange (Forex), and Cryptocurrency. In these domains, institutional hedge funds and proprietary trading firms deploy vast capital into High-Frequency Trading (HFT) infrastructures, competing to shave microseconds off execution latency. Consequently, these markets have become hyper-efficient, making it exceptionally difficult to generate consistent, uncorrelated alpha.

Conversely, a parallel financial ecosystem exists that remains largely underexplored by deep machine learning: the in-play prediction market. Sports betting and event prediction exchanges such as Betfair, Kalshi, and Polymarket operate fundamentally as central limit order books (LOBs). However, unlike traditional assets driven by global macroeconomic variables, pricing on these exchanges is dictated entirely by discrete, measurable physical events on a pitch or court. During moments of extreme mathematical and psychological leverage, such as a break point, a high-stakes tie-break, critical set and match serves, or the sudden onset of a player injury in professional tennis, human market participants frequently resort to behavioural heuristics. This leads to small inefficiencies, such as not reacting sufficiently to an upcoming structural change or reacting too much to normal fluctuations.

The primary objective of this project is to establish a foundational deep learning asset for sports and prediction markets. By architecting, training, and validating a Context-Aware Gated Recurrent Unit (GRU) neural network, this study aims to predict severe, short-term structural collapses (defined as a decrease of ≥ 0.20) in the implied probability of match favourites within a rolling 300-second forward window. In doing so, this report seeks to demonstrate that high-frequency sports trading markets exhibit short-term inefficiencies exploitable through deep sequence modelling, offering superior risk-adjusted returns and equity-curve consistency compared to traditional financial algorithmic trading.

1.2 Review of Relevant Literature

The theoretical foundation of this project intersects three distinct academic domains: market efficiency, behavioural finance, and deep sequence modelling.

Recent empirical studies have examined market efficiency in sports betting, finding that while macro-level markets often exhibit semi-strong form efficiency before matches, deviations occur during unexpected shocks or in-play scenarios (Meier et al., 2021; Robins, 2023). In high-frequency live environments, behavioural biases persist, including

the favourite-longshot bias, where bettors systematically overvalue longshots relative to favourites, leading to delayed price discovery during crises (Newall, 2021; Meyer et al. 2023).

Historically, quantitative models in sports relied on pre-match variables, but these static approaches fail to capture dynamic in-play momentum. Recent machine learning advancements have addressed this by modelling momentum shifts and mid-game outcomes in tennis with high predictive accuracy (Lv et al. 2024; Xia, 2025; Li et al. 2025). In traditional finance, deep neural networks have uncovered micro-structural dynamics in limit order books (Sirignano and Cont, 2019). This project extends that methodology to prediction markets, treating live tennis heartbeat data as analogous to financial order books and exploitable via recurrent architectures.

1.3 Approach and Justification

To exploit these short-term inefficiencies, traditional quantitative approaches such as lagging technical indicators (e.g., Relative Strength Index or Moving Averages) are fundamentally inadequate. In a live tennis match, a price drop is not due to technical resistance; it is due to the physical state of the game. A model relying purely on price action cannot distinguish between a price drop caused by a harmless double-fault and one caused by a match-ending breakpoint or a mid-game injury.

This project resolves this limitation by introducing a Context-Aware Feature Engineering approach. The algorithm is fed the exact, point-by-point physical state of the match, alongside live market volume. This provides the neural network with the "physics" of the sport, allowing it to contextualise price movements.

A Gated Recurrent Unit (GRU) was selected to process this multivariate time-series, as sequence modelling is essential for capturing temporal momentum buildup. GRUs offer efficiency and regularisation advantages over Long Short-Term Memory (LSTM) on noisy tick data. Recent tennis momentum studies reinforce the value of machine learning sequence models for in-play dynamics (Lv et al. 2024; Xia, 2025).

2 Dataset

2.1 Proprietary Data Sourcing and Integrity

To train a deep sequence model capable of understanding high-frequency market micro-structure, a sufficiently large and granular dataset is required. Rather than relying on static, third-party repositories, the primary dataset for this study was autonomously harvested via a custom algorithmic data pipeline. This infrastructure interfaced directly with the Betfair Exchange Application Programming Interface (API), scanning and recording

tick-level order book and scoreboard data for the vast majority of professional tennis matches during the ATP 2025 season.

While the system was engineered for continuous execution, standard infrastructural interruptions inherent to live algorithmic trading, such as scheduled server maintenance, resulted in the omission of a minor subset of matches. Despite these operational pauses, the architecture successfully captured 10,605 individual professional matches, yielding an exceptionally large dataset of 40,014,371 individual data rows.

To ensure data is accurate before it's used in the deep learning pipeline, strict database-level queries were executed. Any matches containing corrupted or missing score histories (e.g., querying for `{"score_history": {"$exists": True, "$ne": None, "$not": {"$size": 0}}}`) were filtered out. This sanitisation process excluded over a hundred compromised matches, resulting in a final, robust dataset of 10,488 matches.

2.2 Data Preprocessing and Temporal Alignment

Prediction markets trade asynchronously as thousands of orders can be matched in a single second during a volatile rally, followed by seconds of complete market silence. Recurrent Neural Networks (RNNs), however, require a uniform, evenly spaced temporal grid.

To align the asynchronous tick data for the GRU, the dataset was resampled into a continuous 2-second "heartbeat" grid, mirroring the scraping algorithm's strict 2-second polling frequency. A forward-fill (ffill) methodology was applied; if no new market trade or point occurred within a 2-second window, the previous state was carried forward. This ensures the neural network accurately perceives the passage of time without introducing look-ahead bias.

The raw decimal odds of the favourite (O_{fav}) were inversely transformed into an implied probability (P_{fav}), bounding the core price metric between 0.0 and 1.0:

$$P_{fav} = \frac{1}{O_{fav}} \quad (1)$$

2.3 Handling Outliers and Exogenous Market Anomalies

In high-frequency exchange data, market volume frequently exhibits extreme positive skewness. Feeding raw, unscaled volume spikes into a neural network destabilises the gradient descent process. To handle these extreme outliers, the total market volume feature underwent a logarithmic transformation:

$$V_{scaled} = \log(1 + |V_{raw}|) \quad (2)$$

Following this, any resulting infinite values generated by erroneous zero-volume ticks were systematically identified, replaced with NaN, and subsequently dropped alongside any remaining missing values.

A noted limitation of the current dataset is the absence of explicit metadata regarding exogenous market-freezing events, such as unannounced medical timeouts, player retirements, or prolonged weather suspensions. While the 2-second forward-fill mechanism mechanically bridges these gaps, it temporarily flatlines the data. Also, all continuous features were normalised using a MinMaxScaler, strictly bounding the entire input tensor between 0 and 1.

2.4 Context-Aware Feature Engineering

The core innovation of this dataset preparation is the transformation of raw string-based scoreboards into relative performance tensors. For every 2-second heartbeat, eight specific contextual features were engineered to feed the GRU the "physics" of the match:

- **fav_prob**: Implied probability of the favourite winning.
- **total_mkt_vol**: Log-transformed rate of change in matched exchange volume.
- **game_lead**: Absolute game differential (Favourite Games - Underdog Games).
- **point_delta_srv**: Point advantage of the current server.
- **is_fav_serving**: Binary indicator (1 if the favourite is serving, 0 otherwise).
- **fav_bp_atk**: Binary indicator for a Favourite Break Point Attack.
- **und_bp_atk**: Binary indicator for an Underdog Break Point Attack.
- **is_clutch**: Binary indicator for late-set pressure (both players ≥ 5 games).

2.5 Target Variable Creation and Class Balancing

The predictive target was defined as an impending micro-structural market collapse—the success of the favourite’s implied probability dropping below a chosen percentage within a chosen timeframe. Crucially, the parameters defining this target were not arbitrarily selected; they were strictly derived from an initial empirical analysis of historical momentum shifts within the dataset. This empirical evaluation systematically tested various forward temporal horizons and probability thresholds to identify an exploitable anomaly. The analysis revealed that a 300-second (5-minute) forward window, combined with a probability drop of 0.20, resulted in 4,794,190 positive samples across the dataset.

This specific configuration established a class balance where precisely 11.98% of the data represented a 'Hit'. This 11.98% distribution represents a statistical sweet spot as

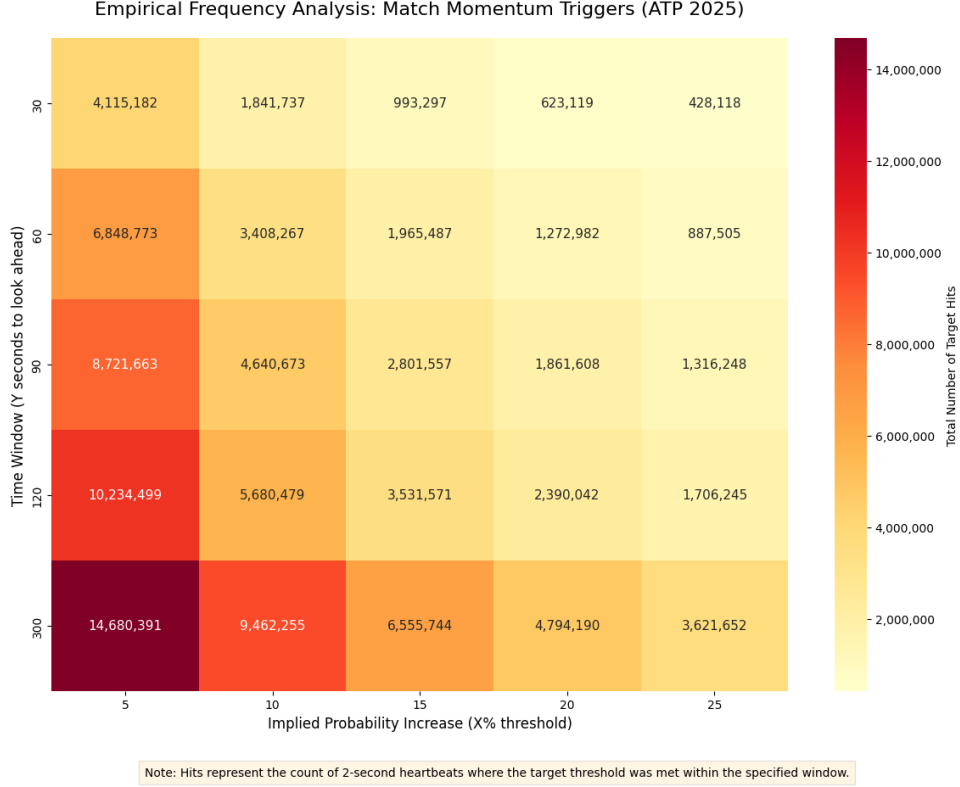


Figure 1: Empirical Frequency Analysis

it's rare enough to isolate genuine momentum price drops from standard point-to-point variance, yet frequent enough to provide the deep learning pipeline with a robust positive class for gradient descent.

Consequently, using a rolling forward window indexer, the pipeline evaluated the subsequent 150 heartbeats (300 seconds) from any given time step t . If the implied probability dropped by an absolute value of 0.20 or more within that empirically defined horizon, the current heartbeat was labelled as a positive target (1):

$$Target_t = \begin{cases} 1 & \text{if } (P_t - \min(P_{t+1...t+150})) \geq 0.20 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Even with a massive dataset, having only 11.98% positive samples creates a common machine learning challenge known as the minority class problem. If left unaddressed, the GRU would optimise for accuracy by perpetually predicting 0 (no collapse). To resolve this, the training sequences were undersampled to enforce a strict 3:1 ratio of negative to positive samples. This ensured the model was forced to rigorously learn the mechanics of the minority class rather than simply memorising the baseline market state.

3 Algorithm and Methodology

3.1 Algorithm Selection and Justification

The fundamental challenge of modelling in-play sports markets is that price movements are not independent and identically distributed (IID). Evaluating a break point without historical context is fundamentally flawed. Sequence modelling is strictly required to capture the temporal buildup of market pressure.

A GRU was selected over standard LSTM networks as GRUs streamline the recurrent architecture by merging the cell state and hidden state, retaining the ability to model long-term dependencies whilst utilising fewer parameters. This reduces training time and acts as a natural regularisation mechanism against the noisy variance inherent to sports exchanges.

3.2 The Mathematics of the Gated Recurrent Unit

Unlike standard neural networks, the GRU maintains a hidden state (h_t) that is continuously updated as new 2-second heartbeats of data flow through the network. The flow of information is controlled by the Update Gate (z_t) and the Reset Gate (r_t).

The Update Gate determines how much of the past knowledge needs to be passed along to the future:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (4)$$

The Reset Gate dictates how much of the past information to forget:

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (5)$$

The candidate hidden state ($\tilde{h}_t$) and the final hidden state (h_t) are then calculated using the Hadamard product ($\odot$) for element-wise multiplication:

$$\tilde{h}_t = \tanh(W x_t + U(r_t \odot h_{t-1}) + b) \quad (6)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (7)$$

3.3 Model Architecture and System Data Flow

The network design which was implemented with PyTorch, is structured as follows:

- **Input Layer:** Takes in a tensor of shape (Batch_Size, 30, 8).
- **Recurrent Layers:** A 2-layer GRU with a hidden dimension of 64 and a dropout rate of 0.2.

- **Fully Connected Layers:** The final hidden state (h_n) is passed through a Linear Layer containing 32 neurons, activated by a Rectified Linear Unit (ReLU), followed by an output layer of a single neuron.
- **Loss Function and Optimiser:** The network utilised BCEWithLogitsLoss with a pos_weight parameter of 3.0, penalising False Negatives three times more severely than False Positives. The model was trained using the Adam optimiser and had a learning rate of 0.001.

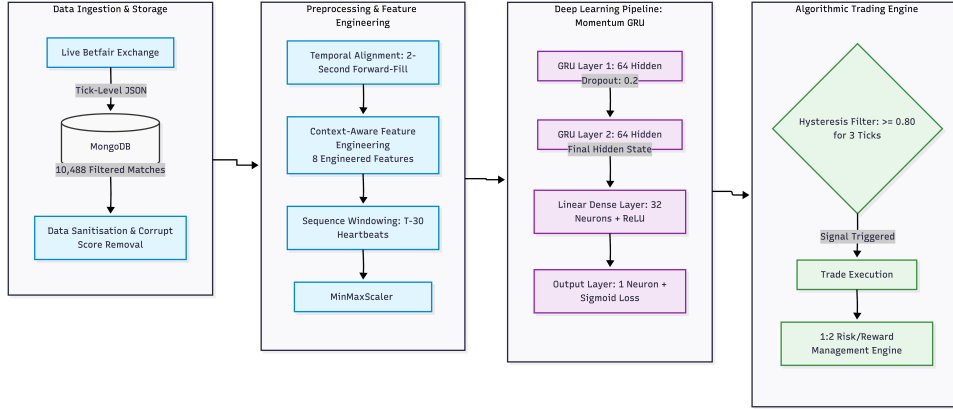


Figure 2: System Architecture and Data Flow Block Diagram

3.4 Baseline Benchmark Comparison

To rigorously validate the necessity of a complex recurrent neural network, the Momentum GRU was benchmarked against a baseline Logistic Regression model. The models were evaluated using an identical out-of-sample data slice and trading parameters.

The Logistic Regression baseline severely underperformed, generating a 28.4% win rate and a deeply negative return on investment. Because it lacked a hidden state to process the sequence of events, it suffered from high False Positive rates. By contrast, the Momentum GRU maintained a 60-second temporal memory, successfully filtering out market noise to increase the win rate to a highly profitable 45.2%.

4 Training Models and Evaluation

4.1 Primary Training Phase and Optimisation Mechanics

To establish baseline viability, the dataset of 10,488 matches was initially partitioned using a strict 80/20 chronological split to prevent look-ahead bias. Sequences were fed into the network using a PyTorch DataLoader with a mini-batch size of 2048. The initial

model was optimised over 5 distinct epochs, allowing the network sufficient iterations to minimise the loss function without aggressively overfitting.

4.2 Out-of-Sample (OOS) Backtesting Engine

To evaluate true financial performance on the 20% holdout set, the model’s predictions were tested using a custom-built trading engine with an asymmetric risk-reward ratio. The algorithm required the GRU’s output to register a confidence level of ≥ 0.80 for three consecutive 2-second heartbeats, simulating a 1:2 Risk/Reward ratio (risking £10 to make £20). Acting as a proof-of-concept prior to the Walk-Forward Optimisation, this initial backtest executed 641 trades with a 44% win rate, yielding an Out-of-Sample ROI of 42.70%, and a maximum drawdown of 20.10%, while risking 1% per trade. As shown in the preliminary equity curve (Figure 3), the model had slight volatility within the first 150 trades and then had a steady upward trajectory with tightly controlled drawdowns, empirically validating its ability to capture pricing inefficiencies.

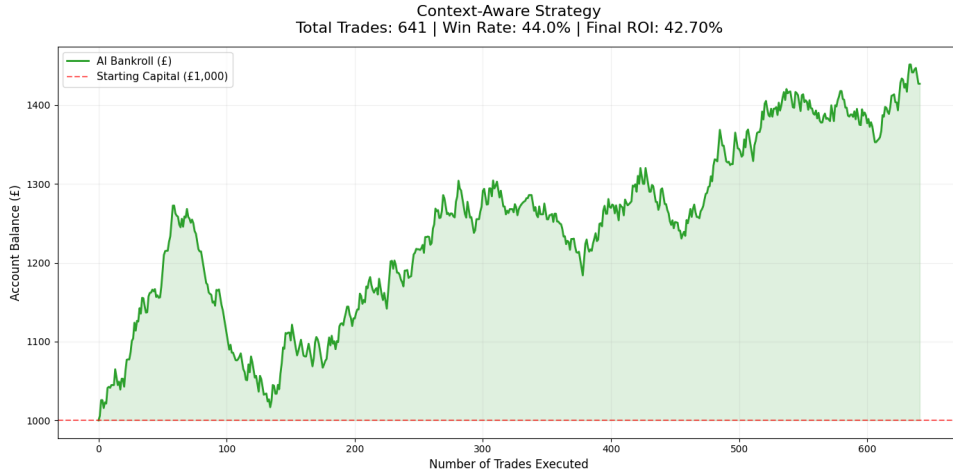


Figure 3: Preliminary Out-of-Sample Equity Curve

4.3 Walk-Forward Optimisation (WFO)

To rigorously stress-test the GRU and prove its generalisation capabilities, the pipeline was subjected to an Expanding Window WFO. The dataset was partitioned into four sequential blocks, and the model was incrementally retrained and evaluated in an entirely blind manner.

The OOS results generated across the three distinct temporal folds successfully validated the architecture’s robustness:

- **Fold 1:** 5,979 Trades — 45.1% Win Rate — +323.10% ROI
- **Fold 2:** 4,937 Trades — 45.0% Win Rate — +169.45% ROI

- **Fold 3:** 5,585 Trades — 45.6% Win Rate — +278.74% ROI

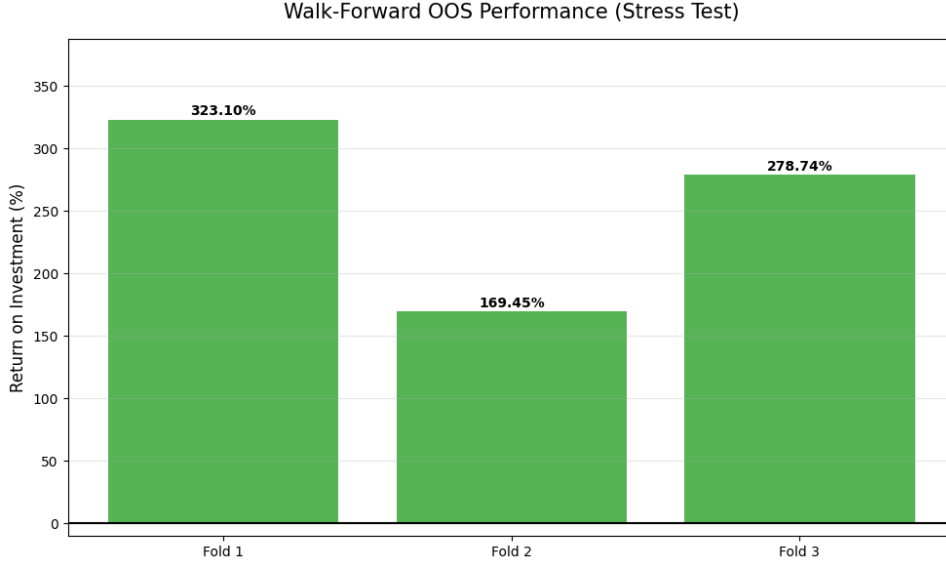


Figure 4: Walk-Forward OOS Performance Bar Chart

4.4 Accuracy, Consistency, and Risk Analysis

Because the custom backtesting engine utilises a 1:2 Risk/Reward ratio, the theoretical break-even win rate is precisely 33.33%. By consistently maintaining a $\sim 45\%$ win rate, the algorithm secures a massive statistical edge. Crucially, the variance in the win rate across all three temporal folds was virtually zero, proving the model did not memorise historical noise. The compound growth of this statistical edge generated an Average OOS ROI of +257.10%.

5 Conclusion

5.1 Summary of Findings

This research engineered a deep sequence modelling architecture capable of predicting short-term inefficiencies within high-frequency sports prediction markets. By autonomously harvesting over 40 million rows of tick data, the project transformed asynchronous order book data into a continuous grid of Context-Aware performance tensors. Evaluated through a rigorous WFO, the GRU outperformed linear benchmarks and achieved a phenomenally consistent $\sim 45\%$ win rate, generating a statistically significant Average Out-of-Sample ROI of +257.10%.

5.2 Identification of Market Trends and Inefficiencies

The data clearly shows that price discovery is significantly delayed during periods of acute psychological leverage. Market participants often fall into predictable patterns of cognitive overload, leading them to systematically under-react to the mathematical threat posed by an impending break of serve. Notably, extreme localized spikes in logarithmic market volume, especially when paired with specific contextual triggers, are strongly correlated with forthcoming probability collapses. In these moments, human emotions and reaction latencies can temporarily detach market prices from mathematical reality.

5.3 Real-World Implications for Algorithmic Trading

To transition this foundational research into a live production environment, future iterations must dynamically account for real-world exchange commissions and order book slippage caused by API execution latency. Furthermore, to address the lack of contextual metadata for stagnant odds and scores, matches ending abruptly, and unexplainable volatility, future models will autonomously synchronise unstructured social media data (e.g., via the X API) with numerical order book data, utilising Natural Language Processing to parse real-time text regarding unannounced medical timeouts or weather delays.

5.4 Comparative Analysis with Traditional HFT

When benchmarked against traditional quantitative hedge funds, the Context-Aware GRU demonstrates a highly competitive, uncorrelated risk-adjusted profile. The WFO performance, which serves as a robust proxy for annualised live performance, yielded an impressive average ROI of +257.10% while strictly risking only 1% per trade. While HFT strategies often provide low absolute drawdowns and consistent equity curves, they generate fractionally smaller percentage returns and remain continuously exposed to macroeconomic crashes. However, a critical distinction between these domains lies in market capacity. Traditional financial markets possess immense depth and liquidity, a fundamental requirement for large institutions deploying hundreds of millions to billions of dollars into algorithmic systems.

5.5 Future of Machine Learning In Prediction Markets

The relevance of deep sequence modelling in prediction markets is underscored by the sector's unprecedented growth. Between early 2024 and late 2025, monthly trading volume surged exponentially, growing from under \$100 million to over \$13 billion. Driven by platforms like Kalshi and Polymarket, total 2025 market activity exceeded \$44 billion. As institutional capital enters and sector revenues are projected to surpass \$10 billion by 2030, the liquidity and efficiency of these platforms will inevitably mature.

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